

Continuum Mechanics

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Chapter 05

Balance Laws
Part 1



Mass and Momentum Conservation

Definition (Mass and Momentum)

Let $\boldsymbol{\phi} : B \times [0, \infty) \to \mathbb{E}^3$ be a motion, with $B_t = \boldsymbol{\phi}_t(B)$, mass density $\rho : B_t \times [0, \infty) \to \mathbb{R}$, and spatial velocity field $\mathbf{v}(\mathbf{x}, t)$. For arbitrary $\Omega_t \subseteq B_t$, the *mass* is

$$\text{mass}[\Omega_t] := \int_{\Omega_t} \rho(\mathbf{x}, t) \, \mathrm{d}V_{\mathbf{x}},$$

and the *linear momentum* is

$$\boldsymbol{\ell}[\Omega_t] := \int_{\Omega_t} \rho(\mathbf{x}, t)\mathbf{v}(\mathbf{x}, t) \, \mathrm{d}V_{\mathbf{x}}.$$



Definition (Angular Momentum)

The *angular momentum* of Ω_t with respect to $\mathbf{z} \in \mathbb{E}^3$ is

$$\mathbf{j}[\Omega_t]_z := \int_{\Omega_t} (\mathbf{x} - \mathbf{z}) \times \left[\rho(\mathbf{x}, t)\mathbf{v}(\mathbf{x}, t)\right] \, \mathrm{d}V_x.$$

Axiom 5.1 (Conservation of Mass)

In the absence of chemical reactions and neglecting relativistic effects, the mass of material undergoing motion in Ω_t does not change over time:

$$\frac{\mathrm{d}}{\mathrm{d}t} \mathsf{mass}[\Omega_t] = 0, \qquad \Omega_t \subseteq B_t, \quad t \in [0, \infty).$$

Definition (Resultant Force and Torque)

Suppose $\Omega_t \subseteq B_t$ is subjected to the body force $\mathbf{b} : B_t \times [0, \infty) \to \mathcal{V}$ and external surface traction $\mathbf{t} : \partial\Omega_t \times [0, \infty) \to \mathcal{V}$. The resultant force is

$$\mathbf{r}[\Omega_t] = \int_{\Omega_t} \rho \mathbf{b} \, \mathrm{d}V_{\mathrm{x}} + \int_{\partial\Omega_t} \mathbf{t} \, \mathrm{d}A_{\mathrm{x}},$$

and the resultant torque about $\mathbf{z} \in \mathbb{E}^3$ is

$$\boldsymbol{\tau}[\Omega_t]_{\mathrm{z}} = \int_{\Omega_t} (\mathbf{x} - \mathbf{z}) \times \rho \mathbf{b} \, \mathrm{d}V_{\mathrm{x}} + \int_{\partial\Omega_t} (\mathbf{x} - \mathbf{z}) \times \mathbf{t} \, \mathrm{d}A_{\mathrm{x}}.$$

Axiom 5.2 (Balance of Momentum)

With respect to a fixed frame of reference, the rate of change of linear momentum of any $\Omega_t \subseteq B_t$ equals the resultant force applied to it. The rate of change of angular momentum about the origin equals the resultant torque about the origin. In symbols,

$$\frac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{\ell}[\Omega_t] = \mathbf{r}[\Omega_t], \qquad \frac{\mathrm{d}}{\mathrm{d}t}\mathbf{j}[\Omega_t]_0 = \boldsymbol{\tau}[\Omega_t]_0, \qquad \Omega_t \subseteq B_t. \tag{5.7}$$

These balances arise from discrete particle systems, where every pair of particles interacts with an energy and every particle is acted upon by an external force. A continuum system may be viewed as a limit of discrete particle systems; see Section 5.1 of the text.



The First Law of Thermodynamics

Definition (Internal Energy)

Let $\boldsymbol{\phi}$ be the motion of a continuum body with mass density ρ and spatial velocity $\mathbf{v}$. For arbitrary $\Omega_t \subseteq B_t$, the *internal energy* is

$$U[\Omega_t] := \int_{\Omega_t} \rho(\mathbf{x}, t)\phi(\mathbf{x}, t) \, \mathrm{d}V_{\mathrm{x}},$$

where $\phi : B_t \times [0, \infty) \to \mathbb{R}$ is the internal energy density per unit mass, or *specific internal energy*.

Here the scalar ϕ is distinct from the bold motion $\boldsymbol{\phi}$.

Definition (Net Heating)

The *net heating* of Ω_t is

$$Q[\Omega_t] := \int_{\Omega_t} \rho(\mathbf{x}, t) r(\mathbf{x}, t) \, \mathrm{d}V_{\mathrm{x}} - \int_{\partial\Omega_t} \mathbf{q}(\mathbf{x}, t) \cdot \mathbf{n}(\mathbf{x}) \, \mathrm{d}A_{\mathrm{x}}. \tag{5.13}$$

Here $r : B_t \times [0, \infty) \to \mathbb{R}$ is the heat supply per unit mass, or *specific heat supply*, and $\mathbf{q} : B_t \times [0, \infty) \to \mathcal{V}$ is the Fourier–Stokes heat flux.

Definition (Power and Kinetic Energy)

The *power of external forces* is

$$\mathcal{P}[\Omega_t] := \int_{\Omega_t} \rho \mathbf{b} \cdot \mathbf{v} \, \mathrm{d}V_{\mathrm{x}} + \int_{\partial\Omega_t} \mathbf{t} \cdot \mathbf{v} \, \mathrm{d}A_{\mathrm{x}}.$$

The *kinetic energy* of Ω_t is

$$K[\Omega_t] := \int_{\Omega_t} \frac{1}{2} \rho(\mathbf{x}, t) |\mathbf{v}(\mathbf{x}, t)|^2 \, \mathrm{d}V_{\mathrm{x}}.$$

The *net working* by external forces is

$$\mathcal{W}[\Omega_t] := \mathcal{P}[\Omega_t] - \frac{\mathrm{d}}{\mathrm{d}t} K[\Omega_t]. \tag{5.14}$$

Remark

If $\mathcal{W}[\Omega_t] = 0$ for $t \in [t_0, t_1]$, then all applied mechanical energy delivered to the continuum body over $[t_0, t_1]$ is used to produce or change the motion.

Axiom 5.3 (The First Law of Thermodynamics)

Considering only thermal and mechanical energy, the rate of change of internal energy equals the sum of net heating and net working:

$$\frac{\mathrm{d}}{\mathrm{d}t} U[\Omega_t] = Q[\Omega_t] + \mathcal{W}[\Omega_t], \qquad \Omega_t \subseteq \mathcal{B}_t.$$

Equivalently, using (5.14),

$$\frac{\mathrm{d}}{\mathrm{d}t} \big(U[\Omega_t] + K[\Omega_t] \big) = Q[\Omega_t] + \mathcal{P}[\Omega_t].$$



Proposition

Suppose the power of external forces can be expressed as

$$\mathcal{P}[\Omega_t] = -\frac{\mathrm{d}}{\mathrm{d}t} G[\Omega_t],$$

where $G[\Omega_t]$ is the potential energy for external forces. Then the First Law is equivalent to

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(U[\Omega_t] + K[\Omega_t] + G[\Omega_t]\right) = Q[\Omega_t].$$

Proof.

($\Rightarrow$): Substitute the assumed expression for $\mathcal{P}[\Omega_t]$ into the equivalent form of Axiom 5.3 and move $-\frac{\mathrm{d}}{\mathrm{d}t} G[\Omega_t]$ to the left-hand side.

($\Leftarrow$): Reverse these algebraic steps. □



Entropy and the Second Law of Thermodynamics

Axiom 5.4 (The Second Law of Thermodynamics)

There exists a quantity called *entropy*,

$$\mathcal{H}[\Omega_t] = \int_{\Omega_t} \rho(\mathbf{x}, t)\eta(\mathbf{x}, t)\, \mathrm{d}V_{\mathrm{x}},$$

where η is the entropy density per unit mass, or specific entropy. There also exists an everywhere-positive temperature field $\theta(\mathbf{x}, t)$ such that the rate of change of entropy minus the heating per unit temperature is nonnegative:

$$\frac{\mathrm{d}}{\mathrm{d}t}\mathcal{H}[\Omega_t] - \left[\int_{\Omega_t} \frac{\rho r}{\theta}\, \mathrm{d}V_{\mathrm{x}} - \int_{\partial\Omega_t} \frac{\mathbf{q}\cdot\mathbf{n}}{\theta}\, \mathrm{d}A_{\mathrm{x}}\right] = \int_{\Omega_t} \sigma(\mathbf{x}, t)\, \mathrm{d}V_{\mathrm{x}} \geq 0,$$

for any $\Omega_t \subseteq B_t$. The field σ is called the *entropy production rate*.

Remark

In the absence of any heating of Ω_t,

$$\frac{\mathrm{d}}{\mathrm{d}t}\mathcal{H}[\Omega_t] = \int_{\Omega_t} \sigma(\mathbf{x}, t)\, \mathrm{d}V_{\mathrm{x}} \geq 0.$$

Thus, when no heating or cooling is applied, the entropy in Ω_t naturally increases.

Because Ω_t is arbitrary, localization yields

$$\sigma(\mathbf{x}, t) \geq 0, \qquad \mathbf{x} \in B_t, \quad t \geq 0.$$

The inequality in Axiom 5.4 is called the Clausius–Duhem inequality.



The Continuity Equation

Result 5.5 (Conservation of Mass)

Let $\boldsymbol{\phi} : B \times [0, \infty) \to \mathbb{E}^3$ be a motion of a continuum body with associated spatial velocity field $\mathbf{v}(\mathbf{x}, t)$ and spatial mass density $\rho(\mathbf{x}, t)$. Then Axiom 5.1 is equivalent to

$$\frac{\partial \rho}{\partial t} + \nabla^{\mathrm{x}} \cdot (\rho \mathbf{v}) = 0, \qquad \mathbf{x} \in B_t, \quad t > 0,$$

or, equivalently,

$$\dot{\rho} + \rho \, \nabla^{\mathrm{x}} \cdot \mathbf{v} = 0, \qquad \mathbf{x} \in B_t, \quad t > 0.$$



Proof.

($\Rightarrow$): Assume Axiom 5.1 holds. Then

$$\text{mass}[\Omega_t] = \text{mass}[\Omega_0] \qquad \text{for every material region and all } t \geq 0.$$

By Result 4.8,

$$\text{mass}[\Omega_t] = \int_{\Omega_t} \rho(\mathbf{x}, t)\, \mathrm{d}V_{\mathrm{x}} = \int_{\Omega} \rho_m(\mathbf{X}, t) J(\mathbf{X}, t)\, \mathrm{d}V_{\mathrm{X}},$$

where

$$\rho_m(\mathbf{X}, t) = \rho(\boldsymbol{\phi}(\mathbf{X}, t), t).$$

Proof (Cont.)

Since $\boldsymbol{\phi}(\mathbf{X}, 0) = \mathbf{X}$,

$$\text{mass}[\Omega_0] = \int_\Omega \rho(\mathbf{X}, 0)\, \mathrm{d}V_\mathrm{X} = \int_\Omega \rho_0(\mathbf{X})\, \mathrm{d}V_\mathrm{X}, \qquad \rho_0(\mathbf{X}) := \rho(\mathbf{X}, 0).$$

Hence the integral form of mass conservation is

$$\int_\Omega \left[\rho_m(\mathbf{X}, t) J(\mathbf{X}, t) - \rho_0(\mathbf{X})\right]\, \mathrm{d}V_\mathrm{X} = 0$$

for every $\Omega \subseteq B$ and every $t \geq 0$. By the Localization Theorem,

$$\rho_m(\mathbf{X}, t) J(\mathbf{X}, t) = \rho_0(\mathbf{X}), \qquad \mathbf{X} \in B, \quad t \geq 0. \tag{5.18}$$

This is the material, or Lagrangian, form of local mass conservation.

Proof (Cont.)

Taking the time derivative of (5.18),

$$\frac{\partial \rho_m}{\partial t}(\mathbf{X}, t) J(\mathbf{X}, t) + \rho_m(\mathbf{X}, t) \frac{\partial J}{\partial t}(\mathbf{X}, t) = 0.$$

Using Result 4.9,

$$\frac{\partial \rho_m}{\partial t}(\mathbf{X}, t) J(\mathbf{X}, t) + \rho_m(\mathbf{X}, t) J(\mathbf{X}, t) \left[\nabla^{\mathrm{x}} \cdot \mathbf{v}\right]_m (\mathbf{X}, t) = 0.$$

Since $J(\mathbf{X}, t) > 0$ for every admissible motion,

$$\frac{\partial \rho_m}{\partial t}(\mathbf{X}, t) + \rho_m(\mathbf{X}, t) \left[\nabla^{\mathrm{x}} \cdot \mathbf{v}\right]_m (\mathbf{X}, t) = 0.$$

Proof (Cont.)

Converting back to a spatial, or Eulerian, description gives

$$\dot{\rho}(\mathbf{x}, t) + \rho(\mathbf{x}, t)\nabla^{\mathrm{x}} \cdot \mathbf{v}(\mathbf{x}, t) = 0.$$

Using Result 4.7,

$$\begin{aligned} 0 &= \frac{\partial \rho}{\partial t}(\mathbf{x}, t) + \nabla^{\mathrm{x}}\rho(\mathbf{x}, t) \cdot \mathbf{v}(\mathbf{x}, t) + \rho(\mathbf{x}, t)\nabla^{\mathrm{x}} \cdot \mathbf{v}(\mathbf{x}, t) \\ &= \frac{\partial \rho}{\partial t}(\mathbf{x}, t) + \nabla^{\mathrm{x}} \cdot \big(\rho(\mathbf{x}, t)\mathbf{v}(\mathbf{x}, t)\big). \end{aligned}$$

($\Leftarrow$): The reverse implication is an exercise. □

Remark

Equation (5.18) will be needed later. The preceding two-line conversion was not essential for the forward direction, because the Reynolds Transport Theorem gives a shorter proof.

Shorter proof.

($\Rightarrow$): Axiom 5.1 and the Reynolds Transport Theorem give

$$
\begin{aligned}
0 &= \frac{\mathrm{d}}{\mathrm{d}t} \int_{\Omega_t} \rho(\mathbf{x}, t) \, \mathrm{d}V_{\mathrm{x}} \\
&= \int_{\Omega_t} \{\dot{\rho} + \rho \nabla^{\mathrm{x}} \cdot \mathbf{v}\} \, \mathrm{d}V_{\mathrm{x}} \\
&= \int_{\Omega_t} \left\{ \frac{\partial \rho}{\partial t} + \nabla^{\mathrm{x}} \rho \cdot \mathbf{v} + \rho \nabla^{\mathrm{x}} \cdot \mathbf{v} \right\} \mathrm{d}V_{\mathrm{x}}.
\end{aligned}
$$

Shorter proof (Cont.)

Combining the last two terms with the divergence product rule gives

$$0 = \int_{\Omega_t} \left\{ \frac{\partial \rho}{\partial t} + \nabla^{\mathrm{x}} \cdot (\rho \mathbf{v}) \right\} \mathrm{d}V_{\mathrm{x}}.$$

Since $\Omega_t \subseteq B_t$ is arbitrary, the Localization Theorem yields

$$\frac{\partial \rho}{\partial t} + \nabla^{\mathrm{x}} \cdot (\rho \mathbf{v}) = 0, \qquad \mathbf{x} \in B_t, \quad t > 0.$$

This equation is called the *continuity equation*. □

Result 5.6 (Transport of Mass-Weighted Fields)

Let $\boldsymbol{\phi} : B \times [0, \infty) \to \mathbb{E}^3$ be a motion with spatial velocity $\mathbf{v}$ and mass density ρ. Let q be any spatial scalar field and let $\Omega_t \subseteq B_t$ be arbitrary. Then

$$\frac{\mathrm{d}}{\mathrm{d}t} \int_{\Omega_t} q(\mathbf{x}, t)\rho(\mathbf{x}, t) \, \mathrm{d}V_{\mathrm{x}} = \int_{\Omega_t} \dot{q}(\mathbf{x}, t)\rho(\mathbf{x}, t) \, \mathrm{d}V_{\mathrm{x}}.$$

Proof.

By the Reynolds Transport Theorem,

$$\begin{aligned}
\frac{\mathrm{d}}{\mathrm{d}t} \int_{\Omega_t} q\rho \, \mathrm{d}V_{\mathrm{x}} &= \int_{\Omega_t} \left[\frac{\partial}{\partial t}(q\rho) + \nabla^{\mathrm{x}}(q\rho) \cdot \mathbf{v} + q\rho \nabla^{\mathrm{x}} \cdot \mathbf{v} \right] \mathrm{d}V_{\mathrm{x}} \\
&= \int_{\Omega_t} \left[\frac{\partial q}{\partial t}\rho + q\frac{\partial \rho}{\partial t} + \nabla^{\mathrm{x}} q \cdot (\rho \mathbf{v}) + q\nabla^{\mathrm{x}}\rho \cdot \mathbf{v} + q\rho \nabla^{\mathrm{x}} \cdot \mathbf{v} \right] \mathrm{d}V_{\mathrm{x}}.
\end{aligned}$$

Proof (Cont.)

Grouping the terms involving q,

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}t}\int_{\Omega_t} q\rho \,\mathrm{d}V_{\mathrm{x}} &= \int_{\Omega_t} \dot{q}\,\rho \,\mathrm{d}V_{\mathrm{x}} + \int_{\Omega_t} q\left[\frac{\partial \rho}{\partial t} + \nabla^{\mathrm{x}}\cdot(\rho\mathbf{v})\right]\mathrm{d}V_{\mathrm{x}} \\ &= \int_{\Omega_t} \dot{q}\,\rho \,\mathrm{d}V_{\mathrm{x}},\end{aligned}$$

using the continuity equation. □

Remark

The proof in the text is more complicated.



Local Conservation of Linear Momentum



Result 5.7 (Local Linear Momentum Balance)

Let $\boldsymbol{\phi} : B \times [0, \infty) \rightarrow \mathbb{E}^3$ be a motion with spatial velocity $\mathbf{v}$ and mass density ρ. Axiom 5.2, balance of linear momentum, is equivalent to

$$\rho \dot{\mathbf{v}} = \nabla^{\mathrm{x}} \cdot S + \rho \mathbf{b}, \qquad \mathbf{x} \in B_t, \quad t > 0,$$

where $S(\mathbf{x}, t)$ is the Cauchy stress field and $\mathbf{b}(\mathbf{x}, t)$ is the spatial body force per unit mass.

Proof.

($\Rightarrow$): Assume the linear-momentum part of Axiom 5.2 holds. Then

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}t}\int_{\Omega_t}\rho\mathbf{v}\,\mathrm{d}V_{\mathrm{x}} &= \int_{\partial\Omega_t}\mathbf{t}\,\mathrm{d}A_{\mathrm{x}}+\int_{\Omega_t}\rho\mathbf{b}\,\mathrm{d}V_{\mathrm{x}}\\ &= \int_{\partial\Omega_t}S\mathbf{n}\,\mathrm{d}A_{\mathrm{x}}+\int_{\Omega_t}\rho\mathbf{b}\,\mathrm{d}V_{\mathrm{x}}\\ &= \int_{\Omega_t}(\nabla^{\mathrm{x}}\cdot S+\rho\mathbf{b})\,\mathrm{d}V_{\mathrm{x}},\end{aligned}$$

using the Divergence Theorem and $\mathbf{t}=S\mathbf{n}$ on $\partial\Omega_t$.

Proof (Cont.)

Using Result 5.6 componentwise,

$$\frac{\mathrm{d}}{\mathrm{d}t}\int_{\Omega_t}\rho\mathbf{v}\,\mathrm{d}V_{\mathrm{x}}=\int_{\Omega_t}\rho\dot{\mathbf{v}}\,\mathrm{d}V_{\mathrm{x}}.$$

Therefore

$$\int_{\Omega_t}(\rho\dot{\mathbf{v}}-\nabla^{\mathrm{x}}\cdot S-\rho\mathbf{b})\,\mathrm{d}V_{\mathrm{x}}=\mathbf{0}.$$

Since $\Omega_t \subseteq B_t$ is arbitrary, the Localization Theorem gives

$$\rho\dot{\mathbf{v}}=\nabla^{\mathrm{x}}\cdot S+\rho\mathbf{b}.$$

($\Leftarrow$): Integrate this local balance over Ω_t, then reverse the preceding steps. □